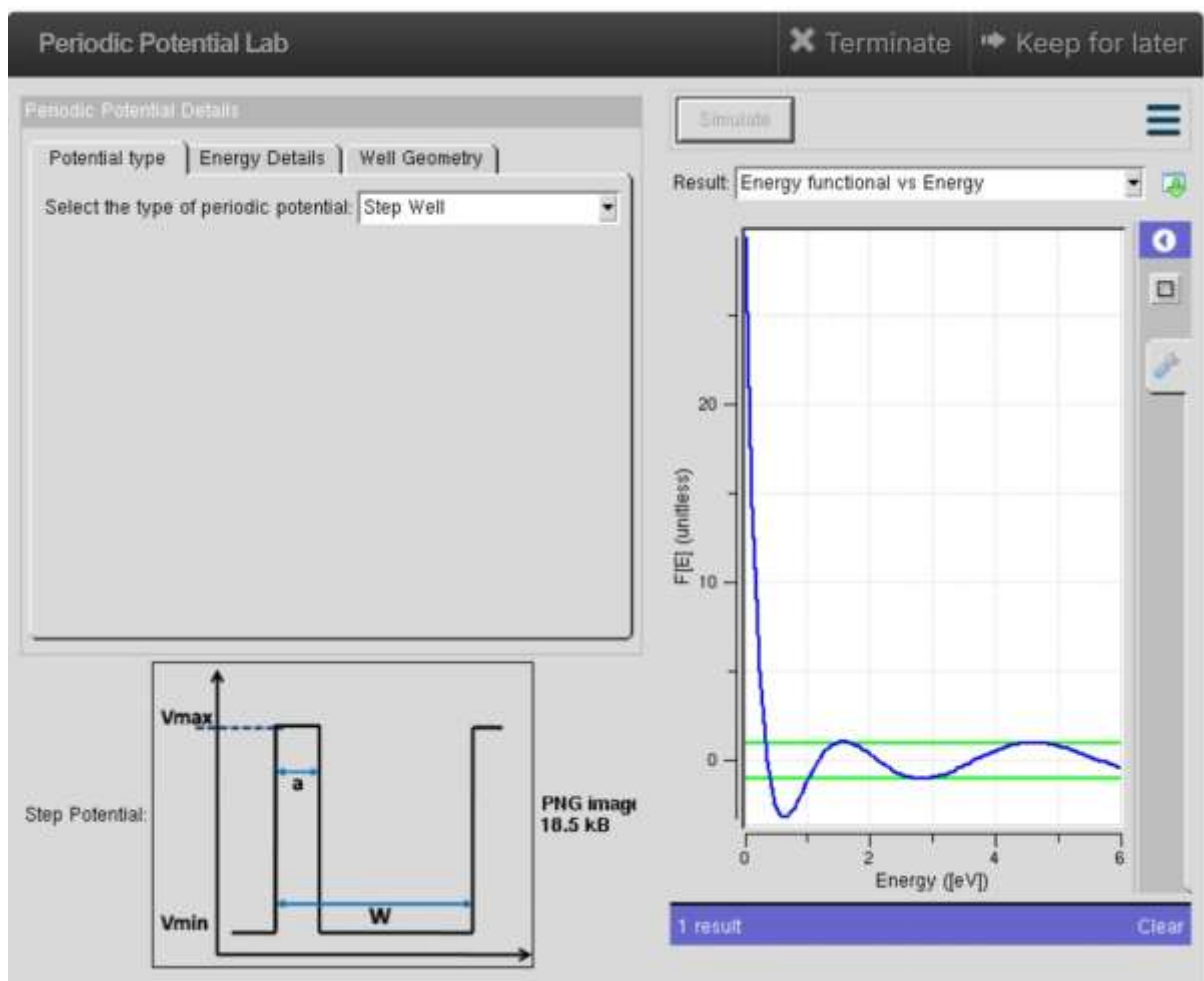


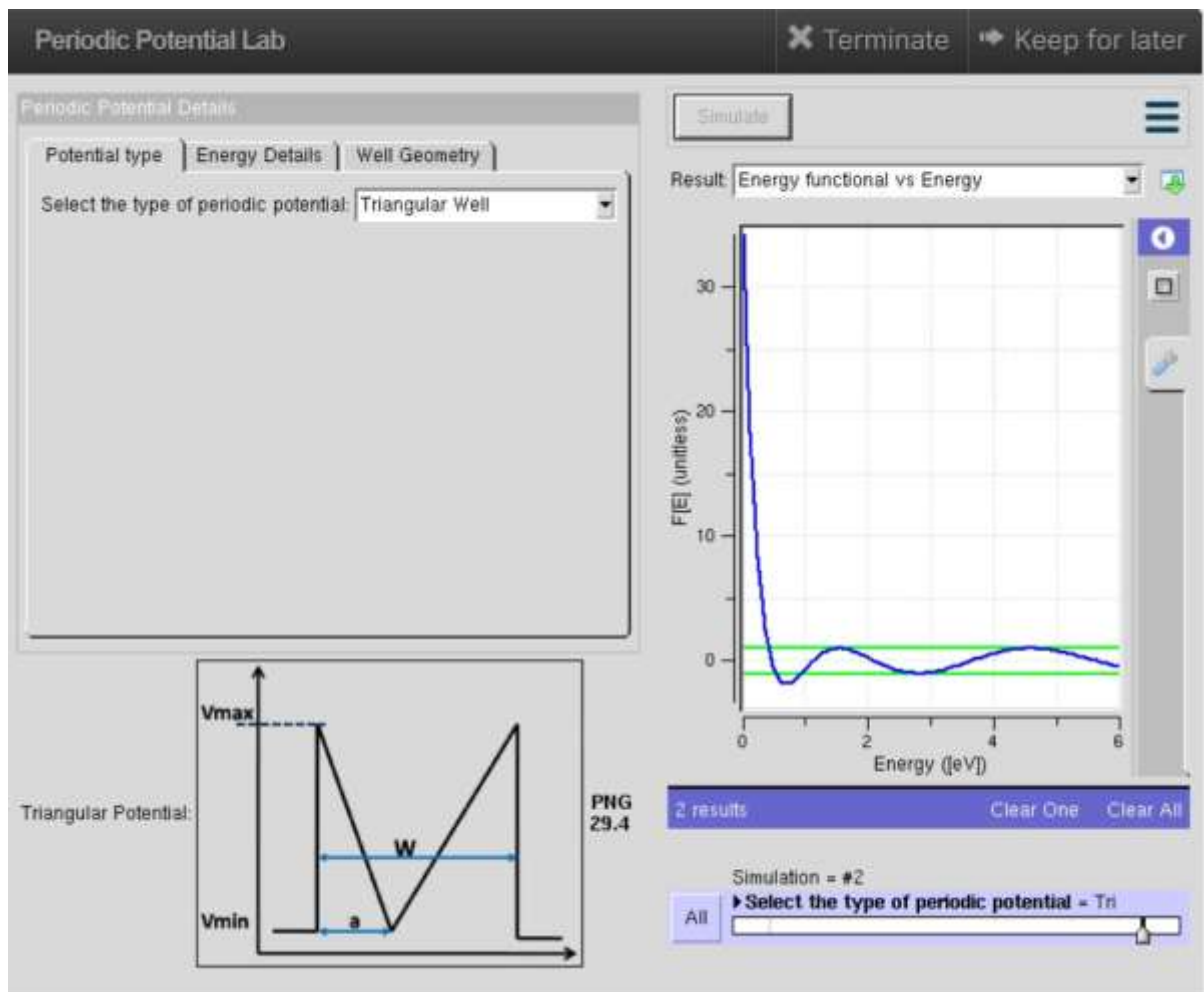
OUTPUT:



This figure shows the **Periodic Potential Lab** simulation using a **Step Well periodic potential**.

The graph illustrates the **Energy Functional vs. Energy**, showing how the electron energy varies within the periodic potential structure.

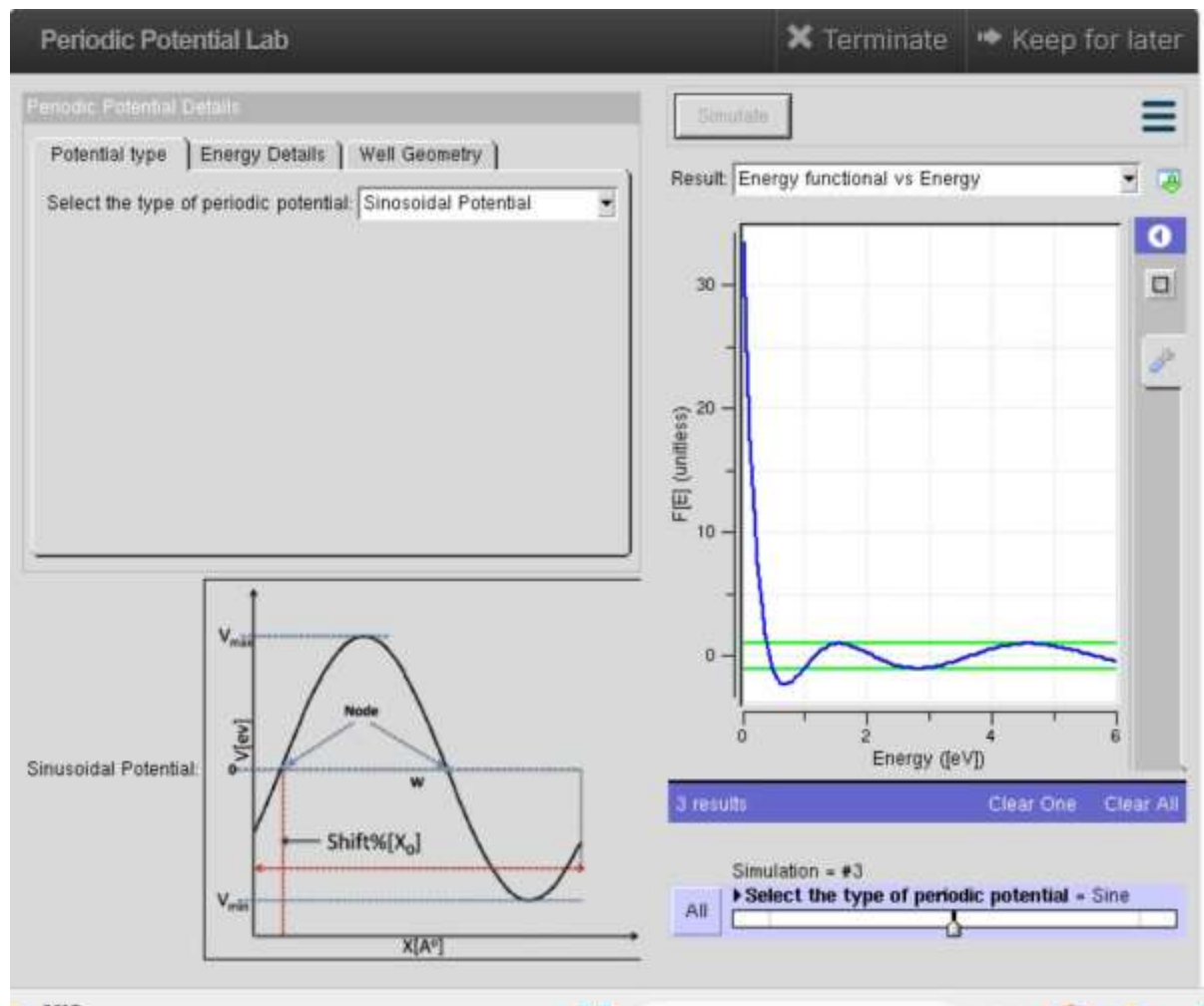
Case Study 1: Simulation of the output for Triangular well



This figure shows the **Periodic Potential Lab** simulation using a **Triangular Well periodic potential**.

The graph shows the **Energy Functional vs. Energy**, illustrating how electron energy varies in the triangular potential well.

Case Study 2: Simulation of the output for Sinusoidal Potential



This figure shows the **Periodic Potential Lab** simulation using a **Sinusoidal Potential**.

The graph shows the **Energy Functional vs. Energy**, illustrating electron behavior in a sinusoidal periodic potential.

Conclusion:

The simulations show that different periodic potential wells (step, triangular, and sinusoidal) affect the electron energy distribution and density of states. These results help analyze and optimize the electronic properties of nanostructured materials for nanoelectronic and thermoelectric applications.